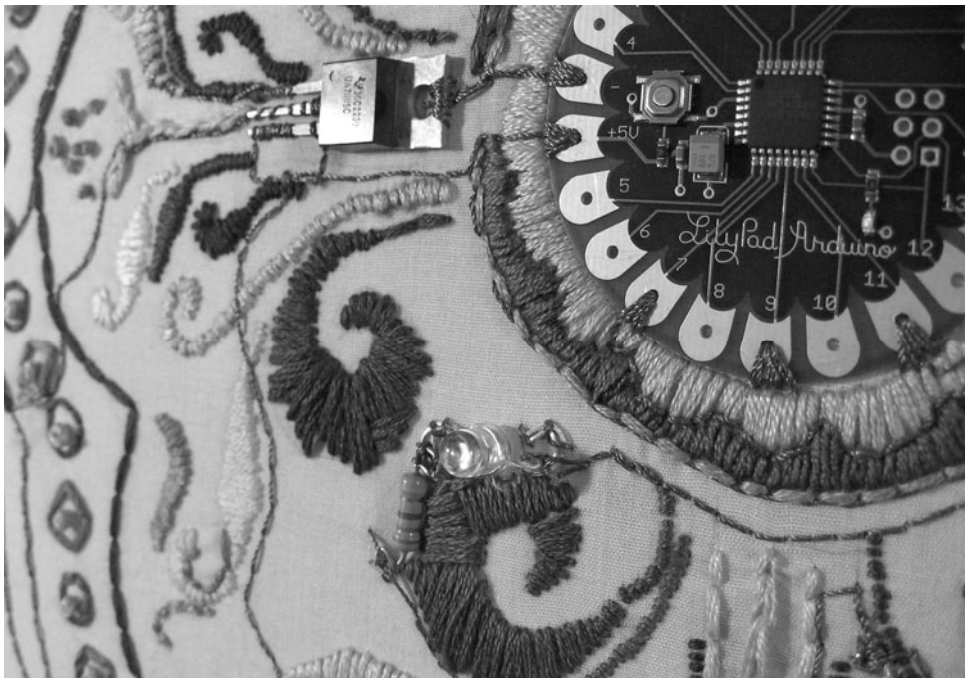


Figure 1-2. *LilyPad Arduino embroidery, courtesy Becky Stern, sternlab.org*

This at least tells us a little bit about what kind of person the Arduino was originally designed for and about the hardware used in its design, but these things alone do not begin to account for the huge degree of success enjoyed by the Arduino as a whole. To get a sense for this popularity, we need to look at the larger Arduino ecosystem and how some fairly divergent parts came together to create a movement.

The Arduino Ecosystem

The Arduino is not just one simple thing making it a little hard to define. It is a microcontroller platform, an open-source design that encourages modification and reuse, a community that has embraced and grown up around the Arduino, and a new crop of projects and devices that can trace their lineage to the Arduino and have in return contributed back to the development of various aspects of the entire Arduino ecosystem.

The Arduino Platform

The Arduino ecosystem begins with the Arduino platform, itself several layers of hardware and software working together to create a cohesive whole. We can start with the hardware interface board—that little, blue circuit board that you build into your projects. It has a fairly standard onboard microcontroller that can interact with the world around it by using its programmable inputs and outputs, as well as a USB